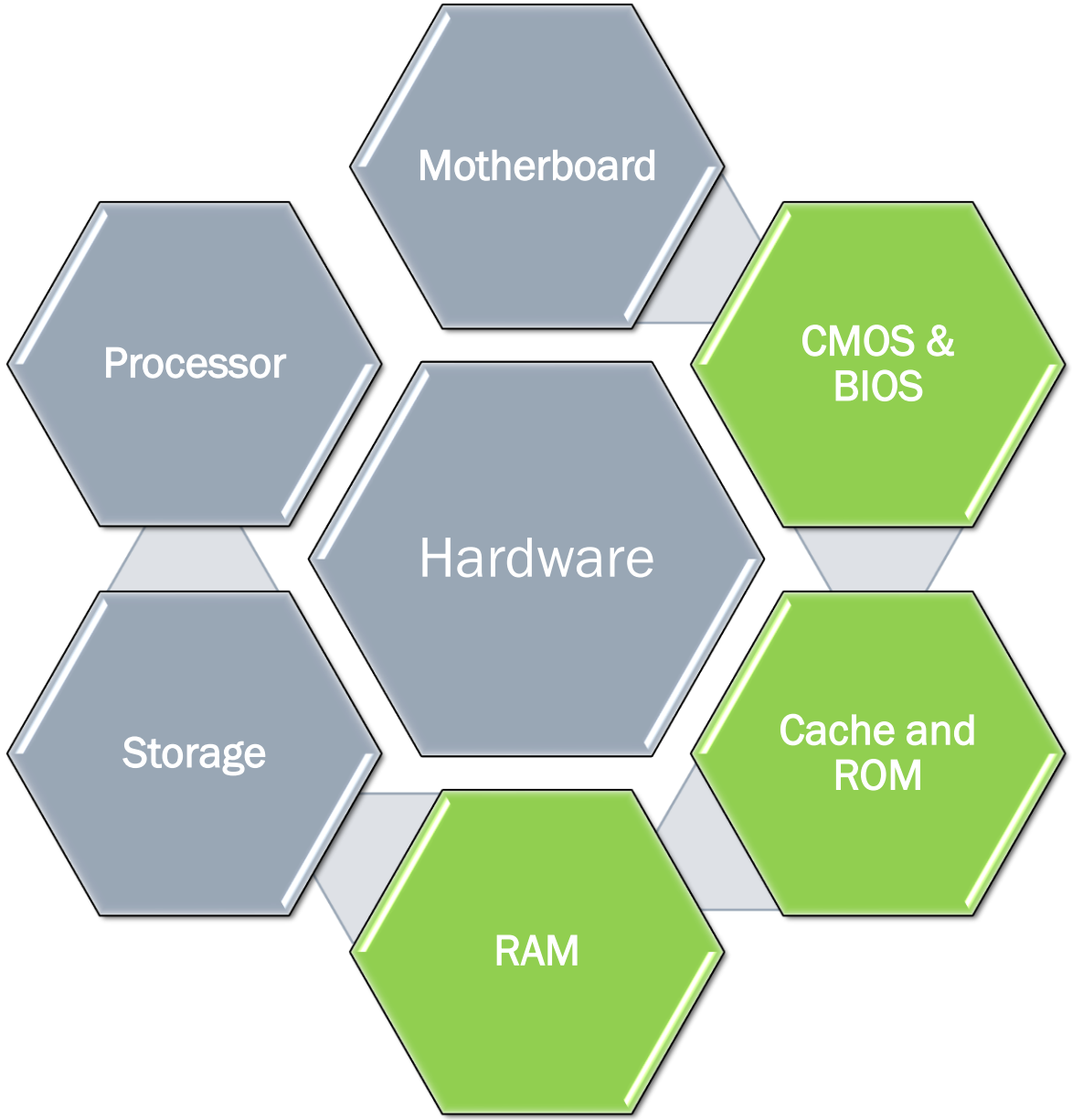
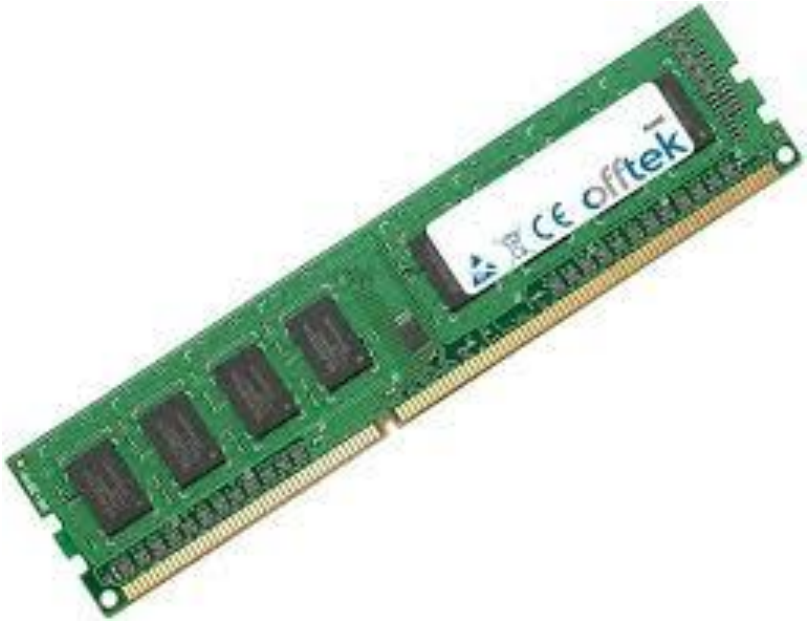

Motherboard
Processor
CMOS &
BIOS
Hardware
Storage
Cache and
ROM
RAM



1- RAM

- **About RAM (Random Access Memory)**

The **RAM (Random Access Memory)** is one of the most important parts of a computer.
It's used to **store temporary information** that your computer needs **right now**.

Basically, the RAM remembers **everything currently running** on your computer —
like when you have multiple browsers open, Photoshop running, and a few folders ready to use —
all of that happens because of **RAM**.

- **How RAM Affects Performance**

How well your applications run **at the same time** depends a lot on the **size of your RAM**.
Accessing data from RAM is **much faster** than accessing data from a hard disk.
Without RAM, your computer would still run applications,
but it would have to load everything directly from the hard disk — which is **much slower**.
So, without enough RAM, running modern software would feel **laggy and slow**.

- **RAM Is Volatile**

RAM is called **volatile memory**,
which means it **loses all data when the power is turned off**.
For example, if you have Photoshop and several browsers open
and you suddenly shut down your computer,
you will **lose everything currently in RAM**,
and you'll have to reopen those programs again later.
That's why it's called *temporary* or *volatile* memory.

▪ How Much RAM Do You Need?

The amount of RAM you need depends on what kind of work you do:

Use Type	Example Activities	Recommended RAM
Basic Use	Browsing, Word, Excel	4–8 GB
Creative Work	Photoshop, Video Editing	8–16 GB
Gaming / Streaming	Modern Games, OBS, 3D Apps	16 GB or more

💡 More RAM allows smoother multitasking and faster performance.

- **Other Factors That Affect Speed**

RAM is not the only thing that determines your computer's speed.

Processor (CPU) and **Graphics Card (GPU)** also play major roles.

Think of it like a team:

•CPU = The brain

•RAM = Short-term memory

•GPU = Visual power

All three must work together to give you the best performance.

- **Types of RAM**

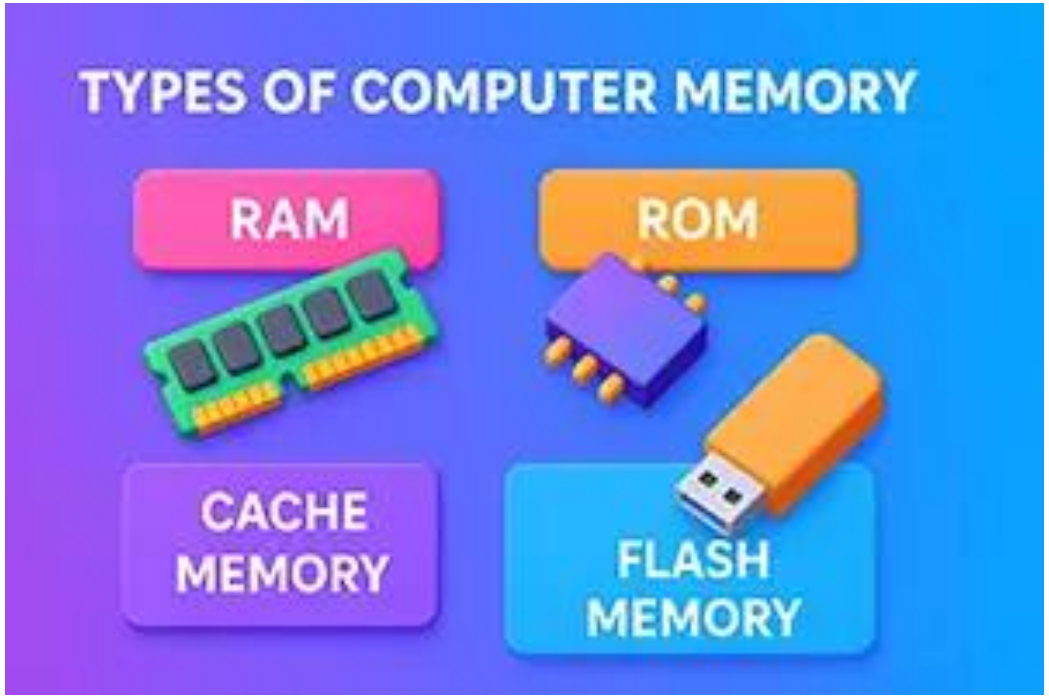
There are different generations of RAM used in computers:

Type	Description
DDR3	Older, slower, used in older PCs/laptops
DDR4	Faster, common in most computers today
DDR5	Newest and fastest, used in high-end systems

💡 DDR stands for "Double Data Rate," meaning data transfers twice per cycle — faster and more efficient.

- **Summary**

•RAM stores temporary data that the CPU needs right now.
•It is faster than a hard disk, but data is lost when the power is off.
•More RAM = better multitasking.
•DDR3, DDR4, DDR5 are the most common types.
•CPU, GPU, and RAM together decide how fast your computer feels.



2- Cache Memory and ROM (Read Only Memory)

- **What Is Cache Memory?**

•Cache memory is a **very fast memory** that sits **between the CPU and the RAM**.
•It **stores the most frequently used data and instructions**.
•This helps the CPU get information **quickly**, without going back to the slower RAM.

💡 **In short:** Cache makes the CPU faster by keeping the “most needed” data nearby.

- **How Cache Memory Works**

1. When the CPU needs data, it **first checks the cache**.
2. If the data is **found in the cache (a cache hit)**, it is loaded immediately.
3. If it's **not found (a cache miss)**, the CPU gets it from the slower **RAM**.

🧠 **Example:**
Imagine you keep your favorite tools in your pocket (cache) instead of your toolbox (RAM).
You save time because you don't have to walk back to get them!

- **Levels of Cache**

There are usually **three types of cache memory**:

Level	Location	Description
L1 Cache	Inside the CPU chip	Very small but fastest.
L2 Cache	Near the CPU (on motherboard or CPU chip)	Slightly larger, a bit slower.
L3 Cache	Shared among CPU cores	Larger but slower than L1/L2.

All three levels work together to keep the CPU running smoothly.

- **Cache Size vs RAM Size**

•Cache memory is **much smaller** than RAM.
•For example:
  - **RAM:** 8 GB
  - **Cache:** A few **MB (megabytes)** only.

•You can see your CPU cache size in **Task Manager** → **Performance** → **CPU tab** (on Windows).

💬 **Why not make cache bigger?**

•Because cache is **very expensive** and **built differently**.
•Making it too large would **slow it down** — it's designed to be **tiny but extremely fast**.

- **Why Cache Memory Is Important**

✅ Speeds up data access for the CPU
✅ Reduces time spent waiting for RAM
✅ Improves performance for repeated tasks
✅ Helps multitasking and overall system speed

💡 Without cache, even a fast CPU would spend more time “waiting” for data.

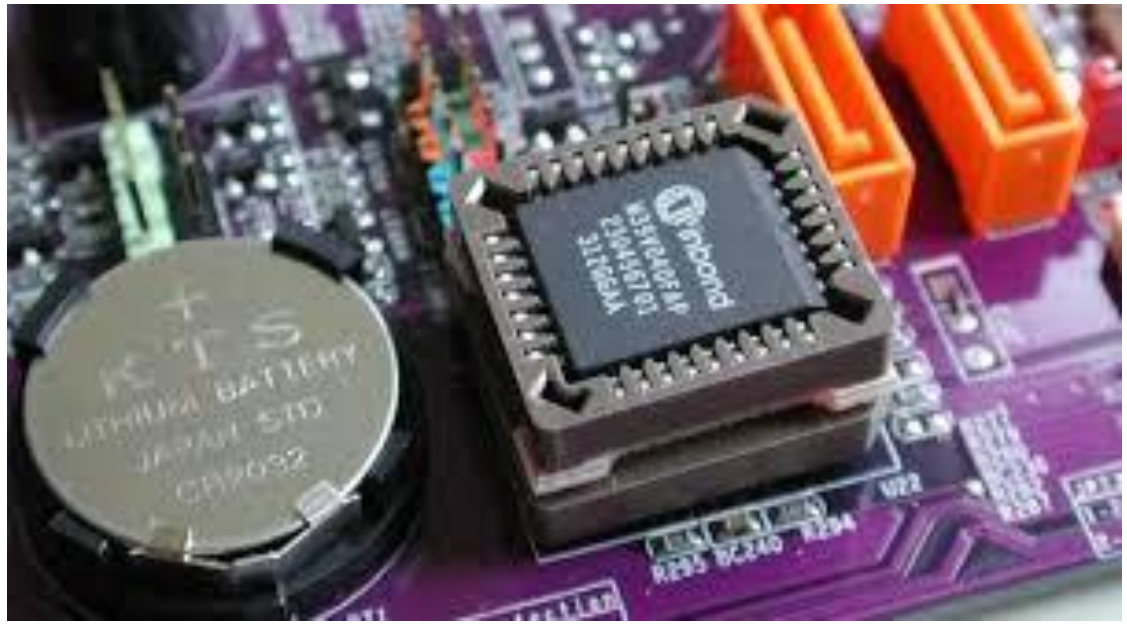
- **What Is ROM (Read Only Memory)?**

•ROM stands for **Read Only Memory**.
•It is a **non-volatile memory** — it **keeps its data even when the computer is off**.
•You **cannot easily modify** the data stored in ROM.

📘 **Example:**
ROM stores the **firmware** that helps your computer **start up (boot)**, such as the **BIOS**.

▪ Features of ROM

Feature	Description
Type of Memory	Non-volatile (keeps data without power)
Main Function	Stores startup instructions and firmware
Data Modification	Very difficult to change
Speed	Slower than RAM or cache
Example	BIOS/UEFI chip on motherboard



CMOS BIOS

3- BIOS and CMOS

What Are BIOS and CMOS?

- Two small but **very important components** inside your computer.
- Both help your computer **start (boot)** and **manage hardware settings**.
- They work **together** every time you press the power button.

💡 Think of them as the *computer's memory and manager* that wake up the system before Windows or macOS even loads.

What Is BIOS?

- **BIOS** stands for **Basic Input/Output System**.
- It is **firmware** (special software stored in a chip on the motherboard).
- It is the **first program** that runs when you turn on your computer.

BIOS = The startup brain of your PC.

What Does the BIOS Do?

- Controls the **first steps of the startup (booting)** process.
- Checks that all hardware (RAM, CPU, keyboard, drives, etc.) is working.
- Loads your **operating system** from the hard drive or USB.
- Provides a menu where you can change **hardware settings**.

⚙️ **Example of BIOS settings you can change:**

- Boot order (which device starts first — hard drive, USB, CD).
- System date and time.
- Fan speed and power management.
- Hard drive or SSD configuration.
- Enabling/disabling built-in hardware (audio, USB ports, etc.).

BIOS Is Non-Volatile

- BIOS settings **do not get lost** when you turn off your computer.
- The BIOS chip keeps its information even when the power is off.
- That's why you don't have to reconfigure your settings every time you start the PC.

💡 *Non-volatile* means the data stays saved even when power is gone.

What Is CMOS?

- **CMOS** stands for **Complementary Metal-Oxide Semiconductor**.
- It's a **tiny memory chip powered by a small battery** on the motherboard.
- It **stores the changes you make** in the BIOS.

🧩 **In simple words:**

BIOS = The program.

CMOS = The small memory that remembers your settings.

How BIOS and CMOS Work Together

1. When you start your computer, **BIOS runs first**.
2. BIOS reads your saved settings (like date, time, and boot order) **from the CMOS chip**.
3. If the CMOS battery dies, the BIOS will **reset to default settings** every time.

🔋 That's why the CMOS battery is important — it keeps your BIOS settings even when the computer is unplugged.

CMOS Battery Example

- Small, round silver battery on your motherboard.
- Usually looks like a coin (called **CR2032 battery**).
- Replacing it is simple and cheap, but it keeps your settings saved.

📸 *If you open a PC, you'll see this small coin-like battery — that's the CMOS battery.*

UEFI – The Modern BIOS

- **UEFI** stands for **Unified Extensible Firmware Interface**.
- It is a **newer and faster version of BIOS**.
- Used in most modern computers.
- It supports:
  - Faster booting
  - Larger hard drives
  - Mouse and graphical interface (instead of old text-only BIOS)

💡 Think of UEFI as **"next-generation BIOS"** — smarter, faster, and more secure.

BIOS vs CMOS vs UEFI

Feature	BIOS	CMOS	UEFI
Type	Firmware (program)	Memory chip	Modern firmware
Location	On motherboard chip	On battery-powered chip	On motherboard
Function	Runs startup & controls hardware	Stores BIOS settings	Replaces BIOS (faster, modern)
Volatile?	No	Yes (needs battery)	No
Example	Boot screen with text	Stores date/time	Modern boot menu with mouse support

- **Summary**

✅ **BIOS** – the system that starts your computer and controls hardware.
✅ **CMOS** – stores your BIOS settings using a small battery.
✅ **UEFI** – the modern version of BIOS with faster startup and advanced features.
✅ Together, they ensure your computer starts correctly every time.

💬 *Without BIOS and CMOS, your computer wouldn't even know how to turn on.*

- 🧠 **Quiz: Computer Memory & System Firmware**

1. What is the main purpose of RAM?
A. Store data permanently
B. Help the CPU run programs faster ✅
C. Display images
D. Connect to the Internet

- **2.** RAM is called **volatile memory** because:
  A. It explodes when hot
  B. It keeps data forever
  C. It loses data when the power is off ✅
  D. It only stores photos
- **3. What does ROM stand for?**
  A. Readable Operation Module
  B. Random Output Memory
  C. Read Only Memory ✅
  D. Regular Operating Machine

4. What is stored in ROM?
A. Games and applications
B. Startup instructions (firmware/BIOS) ✅
C. Temporary files
D. Cache data

5. Which statement about ROM is TRUE?
A. ROM keeps its data even when power is off ✅
B. ROM is used only for videos
C. ROM is faster than cache
D. ROM stores your opened apps

6. What does BIOS stand for?
A. Basic Input/Output System ✅
B. Binary Integrated Operating Software
C. Boot Information Operating Setup
D. Basic Internet Operating Service

7. What is the main job of the BIOS?
A. To play sound
B. To store user passwords
C. To start the computer and load the operating system ✅
D. To connect to Wi-Fi

- **3.** Which statement is TRUE about RAM?
  A. It is slower than a hard drive
  B. It stores data permanently
  C. It is faster than a hard drive ✅
  D. It replaces the CPU
- **4. What happens when you add more RAM to your computer?**
  A. It gets heavier
  B. You can run more programs at the same time ✅
  C. The screen becomes brighter
  D. The keyboard works faster
- **4. Which of the following is a type of RAM?**
  A. DDR4 ✅
  B. SSD
  C. USB
  D. HDD
- **9. Why is cache memory smaller than RAM?**
  A. Because it's old technology
  B. Because it's very expensive and designed to be extremely fast ✅
  C. Because it's slower than RAM
  D. Because it is used only in laptops

5. What does RAM temporarily store?
A. Music and videos
B. Data and programs the CPU is currently using ✅
C. Permanent system files
D. The BIOS firmware

6. Where is cache memory located?
A. Between the CPU and RAM ✅
B. On the hard disk
C. Inside the monitor
D. On a USB drive

7. What is the main purpose of cache memory?
A. To save permanent data
B. To store frequently used data for faster access ✅
C. To store photos and music
D. To back up files online

8. Which cache is the fastest?
A. L1 Cache ✅
B. L2 Cache
C. L3 Cache
D. USB Cache

- **15. Where is the BIOS stored?**
  A. On a chip on the motherboard ✅
  B. In the hard drive
  C. On a USB stick
  D. Inside the CPU
- **16. What does CMOS store?**
  A. Music files
  B. The operating system
  C. BIOS settings such as date, time, and boot order ✅
  D. Temporary internet files
- **17. What powers the CMOS memory?**
  A. Solar panel
  B. A small battery on the motherboard ✅
  C. The monitor
  D. The RAM module
- **18. What happens if the CMOS battery dies?**
  A. BIOS settings reset to default ✅
  B. The CPU stops working
  C. The RAM gets deleted
  D. The system overheats